MyVAM: A Defense-in-Depth Dual-Queue Web Architecture for Deepfake Detection

Noor Adam bin Noor Azmi
Fac. of Info. Science & Technology
Universiti Kebangsaan Malaysia
43600 Bangi, Malaysia
adam.azmi@ukm.edu.my

Siti Norul Huda Sheikh Abdullah
Fac. of Info. Science & Technology
Universiti Kebangsaan Malaysia
43600 Bangi, Malaysia
snhsabdullah@ukm.edu.my

Muhammad Haikal Bin Abdul Razak
Fac. of Info. Science & Technology
Universiti Kebangsaan Malaysia
43600 Bangi, Malaysia
haikalrazakkmm@gmail.com

Ruvenesh Magandaran
Fac. of Info. Science and Technology
Universiti Kebangsaan Malaysia
43600 Bangi, Malaysia
ruvenesh@gmail.com

Ayat Abd-Muti Alrawahneh
Fac. of Info. Science & Technology
Universiti Kebangsaan Malaysia
43600 Bangi, Malaysia
P125852@siswa.ukm.edu.my

Nazhatul Hafizah Kamarudin
Fac. of Info. Science & Technology
Universiti Kebangsaan Malaysia
43600 Bangi, Malaysia
nazhatulhafizah@ukm.edu.my

Sharifah Nurul Asyikin Bt Syed Abdullah
CyberSecurity Malaysia
Level 7 Tower 1, Menara Cyber Axis,
Jalan Impact, 63000 Cyberjaya, Malaysia
sharifah.asyikin@cybersecurity.my

Nik Rafizal Nik Ab Rahim
HLA Integrated Sdn Bhd
B-02-07 & B-02-08, Tamarind Square,
Persiaran Multimedia,
Cyber 10, 63000 Cyberjaya.
nik_rafizal@hla-group.com

Amine Daamouch
CESI École d'Ingénieurs
Campus de Toulouse
16 rue Magellan,
31670 Labège, France
amine.d2amouch@gmail.com

Mohammed Rafique Othman
Fac. of Info. Science & Technology
Universiti Kebangsaan Malaysia
43600 Bangi, Malaysia
rafiqueorton@ukm.edu.my

***Abstract*—The advancement of realistic synthetic media causes threats to digital authentication and information integrity. While advanced deep learning models have shown high laboratory accuracy, they often overlook the architectural complexities required for secure, scalable web deployment. This paper proposes a comprehensive, web-integrated cybersecurity framework for media authentication. Building upon foundational Spatio-Temporal DeepNets, the proposed microservices architecture enforces a defense-in-depth validation architecture utilizing SHA-256-based payload integrity verification, MIMEtype validation, and request-rate limiting to reduce exposure to request-based resource exhaustion. The core forensic engine utilizes a bifurcated pipeline: processing media through a ResNeXt50-backed Bidirectional LSTM alongside a deterministic 8-vector signal heuristic engine. Deployed via a modern Next.js and FastAPI asynchronous stack, the system dynamically routes workloads through a dual-queue Celery architecture (DB0 and DB1) to isolate heavy GPU inference from lightweight heuristic tasks. Load-testing results showed that the asynchronous architecture achieved a 99.8% request success rate and an average API response time of 190 ms for task submission under a simulated load of 100 concurrent users. The architecture also achieved the reported Lighthouse measurements on the web edge (myvam.ukm.my). The hybrid detection configuration produced a preliminary accuracy of 96.4% under the reported evaluation conditions. By closing the gap between theoretical deep learning and practical web engineering, this framework provides a prototype architecture for advances digital forensics platforms.**

***Index Terms*—Web Architecture, Web Security, Deepfake Detection, Temporal Consistency, Microservices, BiLSTM.**

I. Introduction

Deepfake technology is advancing fast and is now a serious cybersecurity threat to identity verification and digital forensics, as attackers increasingly leverage generative adversarial networks (GANs) and diffusion models to bypass biometric checks, fabricate evidence, and run social-engineering campaigns. But there is a big gap between promising AI models and real-world web deployment. Running complex models synchronously can block event loops and be vulnerable to Denial-of-Service (DoS) attacks and standalone defenses can be undermined by adversarial tricks such as noise injection and metadata spoofing when the origin of the media is not verified. To overcome these weaknesses, we propose an asynchronous architecture that combines cryptographic integrity checks, API rate limiting, and multimodal AI analysis. This will treat deepfake detection as an end-to-end web security problem

The primary contributions of this paper are threefold:

1) **Secure Dual-Queue Web Architecture:** We design a decoupled microservices architecture utilizing Next.js,

FastAPI, a dual-queue Redis/Celery broker, and MongoDB. This isolates heavy GPU workloads, reducing API thread-blocking.

2) **Pre-Ingestion Integrity Perimeter:** We introduce an SHA-256 file-integrity mechanism combined with structural metadata analysis (`ffprobe`) as a computationally inexpensive pre-filtering perimeter.
3) **Bifurcated Detection Pipeline:** We deploy a hybrid forensic engine that runs a Spatio-Temporal DeepNet (ResNeXt50 + Bi-LSTM) in parallel with a deterministic 8-vector heuristic engine, providing complementary forensic indicators.

II. Related Works

Research on deepfake detection has focused primarily on deep learning, with CNNs and LSTMs remaining dominant, while practical, scalable web deployment is still limited [1]. Although models such as ResNet50 and ResNeXt50–BiLSTM achieve strong results in spatial and temporal analysis [3], [2], temporal attention methods further improve robustness and interpretability [19]. However, performance may decline when models encounter unseen manipulations, compressed or low-resolution videos, and data distributions that differ from their training sets [6], [7], while hybrid architectures can increase computational costs [9].

Recent systems, including DeepFakeGuard and DeepDect, demonstrate the feasibility of web-based and real-time detection [5], [6]. Several platforms also provide confidence scores, facial crops, frame-level evidence, anomaly indicators, and explainability outputs such as Grad-CAM and LIME [10], [11], [14], [18]. Nevertheless, real-time processing remains difficult because facial localization, feature extraction, temporal modeling, and classification must be completed with minimal delay [5], [13], [9], [15]. Interface functionality alone does not ensure reliability, and excessive concurrent uploads may cause performance degradation or errors [12], [18]; therefore, asynchronous task queues are needed to separate web requests from resource-intensive inference [20].

Security-oriented studies address privacy, traceability, cyberattack resistance, and distributed verification through secure, blockchain-based, and federated-learning frameworks [15], [16], [17], but issues such as request control, queue saturation, resource exhaustion, file validation, and service availability remain insufficiently integrated [21], [4]. As summarized in Table I, this study addresses these gaps through a dual-queue web architecture that separates GPU-intensive ResNeXt50-BiLSTM inference from lightweight heuristic and metadata-processing tasks. Building on the previously developed Decision-Aid Framework, its primary contribution is the practical integration of deterministic forensic analysis, pre-ingestion media validation, request-rate control, and unified task management to improve transparency, scalability, security, and deployment reliability.

III. System Architecture and Threat Model

The proposed framework operates under a formalized threat model wherein sophisticated adversaries attempt to inject manipulated video payloads via web client interfaces to bypass backend authentication servers.

A. Problem Formulation

When processing a continuous stream of user uploads $R = \{r_1, r_2, \ldots, r_n\}$ at time t, each request contains a multimedia payload $P = \{V, M, H\}$, where V denotes the video frames, M the structural metadata, and H the expected SHA-256 digest. Within a constrained web environment defined by a maximum API latency τ_{max} and finite GPU resources C_{gpu}, an adversary may attempt to evade detection by submitting synthetic or manipulated media P_{fake} or exhaust system resources through complex, high-volume uploads that overload the GPU or block the web server's event loop. Thus, web-based deepfake detection can be formulated as a constrained optimization problem that maximizes both detection accuracy, $\mathrm{Acc}(F(P))$, and request success rate, subject to the requirement that the ingestion gateway latency $L_{gateway}$ remain independent of the deep-learning inference time L_{infer}.

B. Microservices Web Architecture and State Management

To mitigate these threats, the system is engineered as a decoupled, asynchronous microservices architecture containerized via Docker. The architecture dynamically balances compute loads using a dual-queue strategy. The stack comprises:

- **Frontend (Client Side):** Built with React and Next.js, with the Web App as the primary interface and a secondary Mobile App for extended access.
- **API Gateway (FastAPI):** Hosted on an ASGI server (Uvicorn) and designed with a dual-mode architecture that uses synchronous processing for rapid data validation and asynchronous processing for task routing.
- **Dual-Queue Message Broker (Redis):** The task queue is split across two isolated Redis databases to prevent resource starvation: heavy tasks are routed to **DB0**, while lightweight tasks are routed to **DB1**.
- **GPU-Isolated Heavy Worker (DB0):** A dedicated Celery worker subscribes exclusively to DB0 and, via Docker resource reservation, is granted pass-through access to a 40GB VRAM GPU instance; it employs dynamic task batching (batch size: 10) to maximize deep learning inference throughput.
- **Lightweight Worker (DB1):** A secondary CPU-only Celery worker subscribes to DB1 to handle lightweight tasks such as deterministic heuristics and metadata extraction.
- **State Management (MongoDB):** Both Celery workers poll their respective queues and securely persist final verdict scores in MongoDB.

TABLE I
COMPARISON OF EVIDENCE-BASED BENEFITS AND LIMITATIONS IN WEB-BASED DEEPFAKE DETECTION

Aspect	Benefits	Limitations
Detection capability	Automated models can detect subtle visual and temporal artifacts that humans are unable to see [4], [5].	Detection accuracy drops significantly when models face new deepfake generation techniques. [6], [7].
Accuracy and modeling	Hybrid models that use spatial and temporal features have shown strong benchmark results, reaching up to 92.5% and 84.06% accuracy in web-based deepfake detection systems [8], [6].	Existing methods still struggle to balance high accuracy with computational efficiency and cross-dataset generalization [9].
Web usability	Web interfaces allow users to upload videos and inspect frame-level outputs with confidence scores. [10], [11].	Does not show reliable classification, as shown by a CNN-based system that incorrectly labeled every test video as fake. [12].
Real-time operation	Support real-time analysis for use cases like live streaming and virtual meeting authentication. [5], [13], [14].	Real-time performance remains difficult, where systems must balance low latency with high detection accuracy. [15], [9].
Security and privacy	Incorporate identity verification, privacy-preserving AI training, and blockchain-based records. [11], [16], [17].	Security is not yet a standard evaluation metric, and system security and latency are often overlooked [4].
Multimodal detection	Multimodal are more robust because they analyze both audio and video rather than only facial frames [15], [7].	Processing audio in multiple languages and maintaining strong performance remain a challenge. [15], [7].
Scalability	Robust systems and scalable data pipelines allow the model to do content filtering, digital forensics, and media verification. [15], [5].	Load testing shows server performance can drop significantly at higher concurrency. [18].
Interpretability	Web apps improve understanding by showing evidence, such as Grad-CAM and LIME explanations, and cropped regions to aid inspection. [11], [18], [14].	Explainability makes inspection easier but does not resolve issues like bias, overconfidence, or dependence on specific datasets [8], [9].

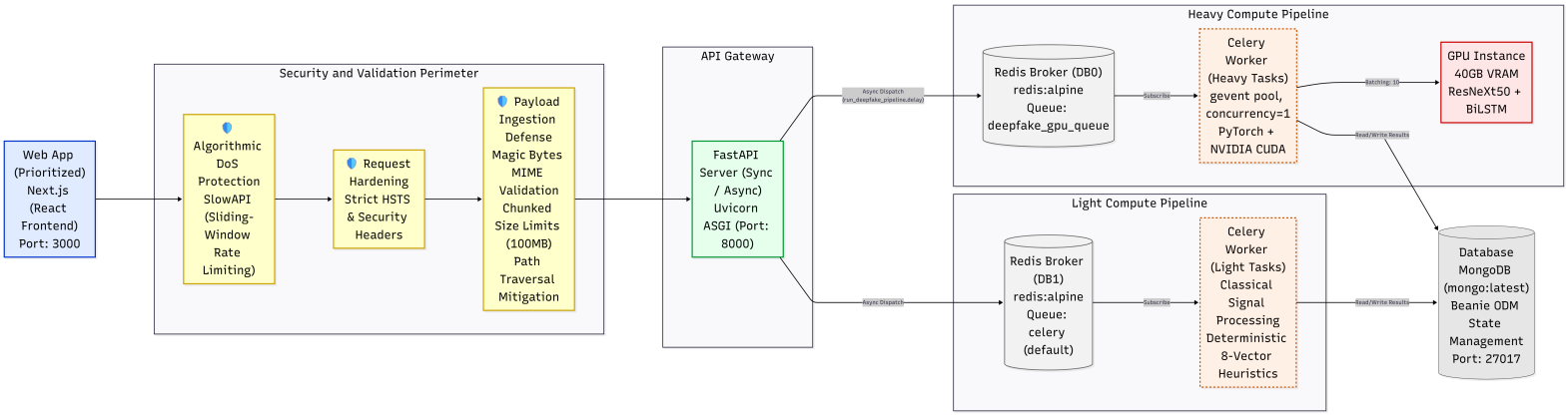


Fig. 1. The Defense-in-Depth Dual-Queue Web Architecture, detailing the security perimeter, FastAPI gateway, and bifurcated Celery worker pipelines.

C. API Security and Algorithmic DoS Mitigation

The FastAPI gateway is able to enforce a rate limiting system for video processing requests from each IP address using an in-memory sliding window. If a user exceeds their allotted number of video processing requests in a specified time frame, they will receive an HTTP 429 Too Many Requests response and be prevented from continuing to send requests to the backend worker. However, distributed production environments would need a centralized rate limiting mechanism.

Algorithm 1 demonstrates the implementation of this algorithmic DoS protection layer in the FastAPI gateway.

D. Pre-Ingestion Integrity and Media-Type Validation

Before entering the processing queues, each media file undergoes type validation and SHA-256 hashing. The resulting digest is compared with the trusted hash stored during media ingestion to detect alterations within the defined submission workflow, providing file-integrity verification without independently confirming the media's authenticity or provenance.

Algorithm 2 shows the validation logic used before data is allowed to enter the Redis queue.

IV. THE SPATIO-TEMPORAL DEEPNET ENGINE

After the uploaded video passes the initial validation, the system gives the AI processing to DB0.

A. Cascading Face Localization

Accurate facial cropping is essential for reducing false positives. Therefore, the architecture uses a cascading pipeline primarily on the Multi-task Cascaded Convolutional Network (MTCNN), with OpenCV Haar Cascade as a fallback when

Algorithm 1 FastAPI middleware for basic sliding-window request-rate limiting

Global: $rateLimit \leftarrow 5$ ▷ *Maximum requests*
Global: $timeWindow \leftarrow 60$ ▷ *Rolling window in seconds*
Global: $clientRequests \leftarrow \{\}$ ▷ *Hash map for IP tracking*

function RATELIMITMIDDLEWARE($request, callNext$)
    $clientIp \leftarrow request.clientIp$
    $currentTime \leftarrow$ GETCURRENTTIME
    ▷ *Sliding-Window Evaluation*
    $reqs \leftarrow$ GETORDEFAULT($clientRequests, clientIp, []$)
    ▷ *Filter requests older than the time window*
    $reqs \leftarrow \{t \in reqs \mid (currentTime - t) < timeWindow\}$
    if LENGTH($reqs$) $\geq rateLimit$ **then**
        return Response(429, "Too Many Requests: Request burst protection active.")
    end if
    APPEND($reqs, currentTime$)
    $clientRequests[clientIp] \leftarrow reqs$
    return CALLNEXT($request$)
end function

Algorithm 2 SHA-256 File-Integrity and Media-Type Validation

function VALIDATESECUREPAYLOAD($filePath, expectedHash$)
    ▷ *Enforce strict MIME-type check to prevent spoofing*
    $mimeType \leftarrow$ EXTRACTMIMETYPE($filePath$)
    $allowedTypes \leftarrow$ {"video/mp4", "video/x-matroska", "video/webm", "video/quicktime", "video/x-msvideo"}
    if $mimeType \notin allowedTypes$ **then**
        raise Exception(415, "Invalid Media Type: Possible Spoofing Attack")
    end if
    ▷ *SHA-256 Integrity Validation*
    $hash \leftarrow$ INITIALIZESHA256
    $file \leftarrow$ OPEN($filePath$, "rb")
    while $file$ has more data **do**
        $byteBlock \leftarrow$ READCHUNK($file, 4096$)
        UPDATE($hash, byteBlock$)
    end while
    CLOSE($file$)
    $calculatedHash \leftarrow$ HEXDIGEST($hash$)
    if $calculatedHash \neq expectedHash$ **then**
        raise Exception(403, "Integrity Hash Mismatch: Payload Tampered")
    end if
    return true
end function

no face is detected. Lastly, the architecture will use center cropping to produce consistent 224 × 224 matrices when previous methods are unable to crop the faces.

B. ResNeXt50 and Bi-LSTM Modeling

Following previous benchmarks [3], a pre-trained `ResNeXt50_32x4d` extracts a spatial feature sequence, F_t, from the cropped faces. To catch glitches like unnatural changes between frames, this sequence is then put into a Bidirectional LSTM (Bi-LSTM) network with a 2048 latent dimension:

$$h_t = [\overrightarrow{\text{LSTM}}(F_t), \overleftarrow{\text{LSTM}}(F_t)] \quad (1)$$

To improve attention stability, the model includes Adaptive Selective Temporal Attention Regularization (ASTAR) [19] and Cross-Temporal Attention Weights (CTAW).

V. Deterministic 8-Vector Heuristic Engine

The proposed system applies signal processing for transparency to overcome classification without any explanation. These lightweight tasks run asynchronously on the DB1 server and provide justifications by analyzing eight features:

1) **Spectral Artifacts (FFT):** Finds Spectral anomalies and irregular frequencies produced by GAN-based upsampling.
2) **DCT Leakage:** Search for abnormal frequency distributions and leakage patterns within the image's cosine transform coefficients.
3) **Boundary Blending:** Applies a Laplacian edge detector to isolate manipulated or poorly blended facial boundaries.
4) **Eye Inconsistency:** Analyzes pixel and texture patterns around the eyes to reveal unnatural characteristics.
5) **Temporal Color Flicker:** Detects color-balance shifts and frame-to-frame inconsistencies across video frames.
6) **JPEG Re-compression:** Computes the Mean Absolute Error (MAE) to determine whether the media has been through multiple JPEG compression cycles.
7) **Sharpness Jitter:** Evaluates temporal fluctuations in Laplacian variance to detect inconsistent sharpness and focus changes throughout the video.
8) **Illumination Asymmetry:** Analyzes variations in the CIELAB color space to find lighting inconsistencies and asymmetric illumination across the face.

VI. System Performance and Evaluation

A. Web Application Interface (MyVAM)

The proposed architecture is implemented in MyVAM (MyVideoAsli Model), a user-friendly web application with Next.js interface that enables forensic analysts to navigate data efficiently. As shown in Figure 2, the web application allows users to upload video files and be able to select specific detection models for analysis.

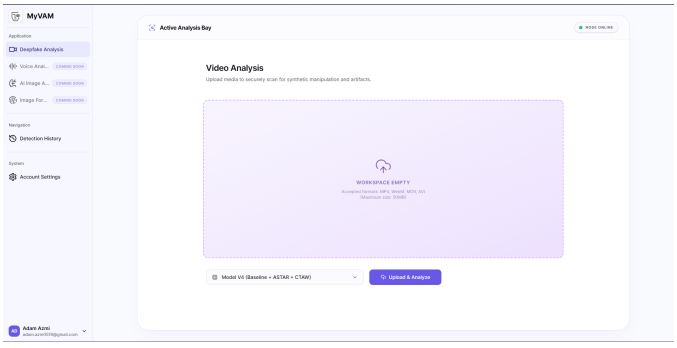


Fig. 2. MyVAM analysis bay, interface for video uploads.

After FastAPI and Celery process the video, the system generates a detailed forensic report, as illustrated in Figures 3 and 4. For the original video, the system classifies the media as "Authentic Media" with 86.7% confidence, reports eight forensic metrics, extracts embedded metadata, and provides

frame-by-frame analysis. In contrast, the manipulated video is classified as "Synthetic Media" with 100.0% confidence, with elevated Temporal Color Flicker (63.1%) and Eye/Blink Inconsistency (58.4%) highlighted as key indicators of manipulation. This transparent analysis gives investigators concrete evidence rather than an unexplained binary prediction.

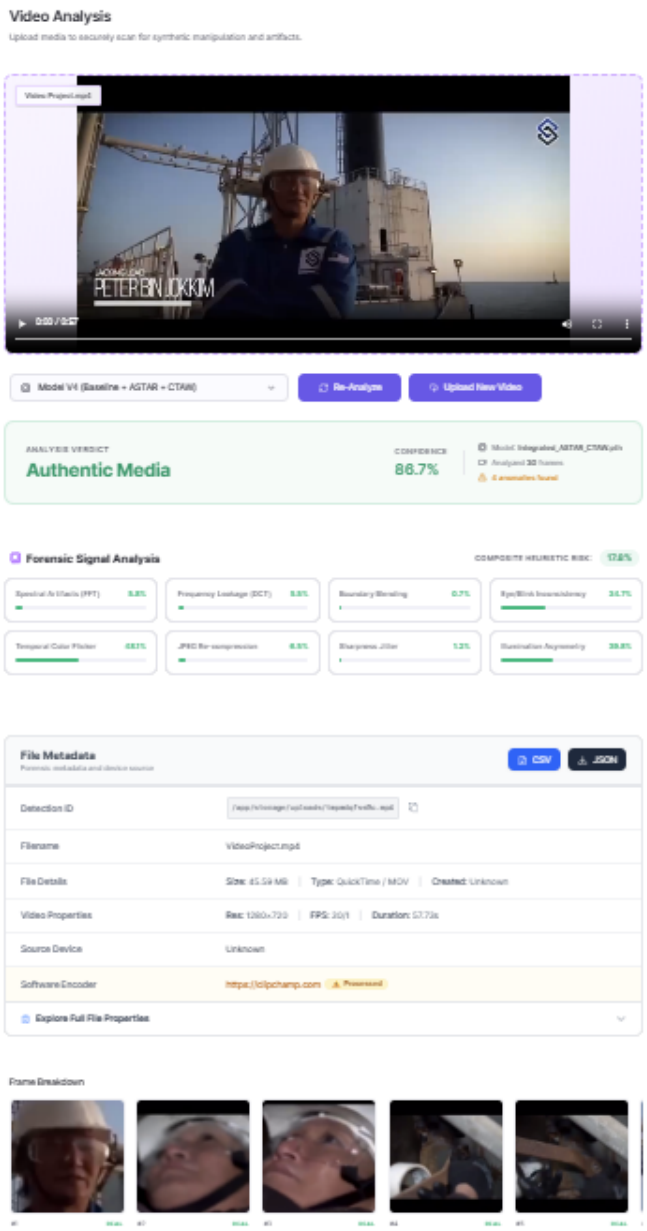


Fig. 3. Analysis report for an authentic video, demonstrating low heuristic risk scores and high confidence.

B. Web Architecture Performance and Lighthouse Scores

To validate the necessity of the separated microservices design, load testing was conducted using Apache JMeter. As shown in Table II, the synchronous baseline suffered catastrophic degradation. In contrast, the dual-queue architecture

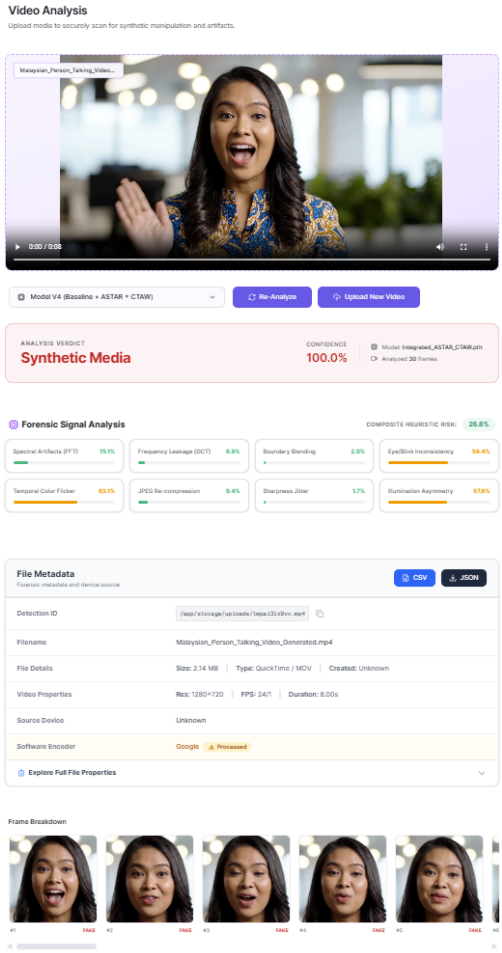


Fig. 4. Analysis report for a deepfake video, highlighting isolated anomalies in Temporal Color Flicker and Eye Inconsistency.

maintained a sub-200ms API response time, maintaining the reported performance under the tested load.

TABLE II
API TASK-SUBMISSION PERFORMANCE UNDER SIMULATED LOAD

Architecture	Users	Avg. API Acknowledgement Time (ms)	Success Rate
Sync. Flask	100	N/A	11.2% (Crash)
Async FastAPI/Celery	100	**190**	**99.8%**

The reported response time measures only the API gateway's request validation and task submission to the asynchronous queue, rather than the complete end-to-end video-analysis duration, which varies according to video length, queue waiting time, face extraction, model inference, and heuristic analysis. To evaluate the prioritized Next.js web application's edge-client performance, a Google Lighthouse audit was conducted on the production domain (`myvam.ukm.my`); as shown in Table III, the decoupled frontend achieved the reported measurements by offloading processing delays to

background Celery brokers.

TABLE III
GOOGLE LIGHTHOUSE WEB PERFORMANCE METRICS FOR MYVAM.UKM.MY

Metric	Result
Performance	94
Accessibility	98
Best Practices	100
SEO	100
First Contentful Paint (FCP)	0.8s
Time to Interactive (TTI)	1.2s
Cumulative Layout Shift (CLS)	0.01

C. Detection Efficacy

Table IV presents preliminary system-level results from a limited prototype evaluation using the FaceForensics++ dataset, which contains 1,000 authentic and 4,000 manipulated videos. Based on a held-out set of 500 test videos compressed with H.264 at CRF 23, a video was classified as synthetic when its DeepNet score exceeded 0.50 or when at least three heuristic indicators surpassed thresholds established at the 95th percentile of the authentic validation scores. The Hybrid System achieved 96.4% baseline accuracy and improved performance under the evaluated compression condition; however, these findings should not be interpreted as evidence of general robustness against all deepfake-generation or adversarial techniques.

TABLE IV
DETECTION PERFORMANCE UNDER THE EVALUATED CONDITIONS

Evaluation Condition	DeepNet Only	Heuristics	Hybrid System
Pristine (No Attack)	94.2%	86.5%	**96.4%**
Compression (H.264)	81.3%	89.1%	**92.8%**

VII. CONCLUSION

This paper presents a prototype dual-queue architecture for web-based deepfake detection that separates GPU-intensive inference from lightweight metadata and heuristic analysis using FastAPI, Celery, Redis, and MongoDB. The system incorporates media-type validation, SHA-256 integrity checking, and API request-rate limiting, achieving a 99.8% request success rate and 190 ms average acknowledgement time under a 100-user load test. Preliminary evaluation yielded hybrid accuracies of 96.4% for pristine media and 92.8% under the tested H.264 compression condition; however, broader testing across datasets, compression levels, adversarial transformations, distributed rate limiting, and end-to-end latency is necessary before drawing general conclusions about its security, scalability, and robustness.

ACKNOWLEDGMENT

This research and system development are supported by the Ministry of Higher Education Malaysia under the research grant for *Deepfake Detection Using Optimized Multi-Stage DeepNet* (Grant Reference: IMAP/3/2024/ICT07/UKM//1).

REFERENCES

[1] A. Abd-Muti Alrawahneh et al., "Video authentication detection using deep learning: a systematic literature review," *Applied Intelligence*, vol. 55, p. 239, 2025.
[2] A. Abd-Muti Alrawahneh et al., "Decision-Aid Framework for Face Authentication Detection Using ResNext50 and BiLSTM to Enhance Media Integrity," *IEEE Access*, vol. 13, 2025.
[3] A. Abd-Muti Alrawahneh et al., "A Comparative Study of Pre-Trained CNN Architectures for Detecting AI-Generated Versus Human-Created Images," *Malaysian Journal of Cybersecurity and Applications*, 2024.
[4] A. Heidari et al., "Deepfake detection using deep learning methods," *WIREs*, 2023.
[5] V. S. R. et al., "DeepFakeGuard: Real-Time Deepfake Video Detection Leveraging Celeb-DF Dataset and CNN-LSTM Framework," *ICOECA*, 2025.
[6] W. X. Tay et al., "DeepDect: A Facial Deepfake Video Detection Application using Ensemble Learning," *ICSPC*, 2024.
[7] M. Alrashoud, "Deepfake video detection methods, approaches, and challenges," *Alexandria Engineering Journal*, vol. 64, 2025.
[8] A. K. Jha et al., "Deep Learning Based Deepfake Video Detection System," *ICDT*, 2025.
[9] A. Kaur et al., "Deepfake video detection: challenges and opportunities," *Artificial Intelligence Review*, vol. 57, 2024.
[10] K. Lee, I. Jung, and S. S. Woo, "iFakeDetector: Real Time Integrated Web-based Deepfake Detection System," *IJCAI*, 2024.
[11] Ranjan M B, "Detection of Face Swapped Deep Fake Videos," *IJSREM*, 2025.
[12] S. Jain, "Deep Fake Detection: Using a web Based Convolutional Neural Network System," *IJSREM*, 2025.
[13] A. Doshi et al., "Realtime Deepfake Detection using Video Vision Transformer," *IBSSC*, 2022.
[14] S. Kumar et al., "Deepfake Detection System Using Hybrid Deep Learning for Images and Videos," *ICAECT*, 2026.
[15] N. Kumar and A. Kundu, "SecureVision: Advanced Cybersecurity Deepfake Detection with Big Data Analytics," *Sensors*, vol. 24, 2024.
[16] A. Heidari et al., "A Novel Blockchain-Based Deepfake Detection Method Using Federated and Deep Learning Models," *Cognitive Computation*, vol. 16, 2024.
[17] M. Wazid et al., "A Secure Deepfake Mitigation Framework: Architecture, Issues, Challenges, and Societal Impact," *Cyber Secur. Appl.*, vol. 2, 2024.
[18] A. Rahmatulloh et al., "Web-Based Deepfake Detection Using VERITAS: Integrating Vision-Based Excitation with Transformer-Driven Intelligence," *Jurnal RESTI*, vol. 10, 2026.
[19] A. Abd-Muti Alrawahneh et al., "A novel temporal attention regularization loss for robust and explainable deepfake detection," *Discover Artificial Intelligence*, 2026.
[20] S. N. Ahmad et al., "A Systematic Review of The Factors and Management Implications of Information Technology (IT) and System Development Project Failure," *Malaysian Journal of Information and Communication Technology (MyJICT)*, vol. 11, no. 1, pp. 1–15, 2026.
[21] S. N. H. S. Abdullah et al., "Evaluating Deep Learning Models for Web-Based Video Forgery Detection," *Asia-Pacific Journal of Information Technology and Multimedia*, vol. 14, 2025.